ICML
2026
REPRO AUDIT

Diffusion Models Are Statistically Optimal
Scoped CPU Audit

CPU audit of intrinsic-dimension diffusion rates.

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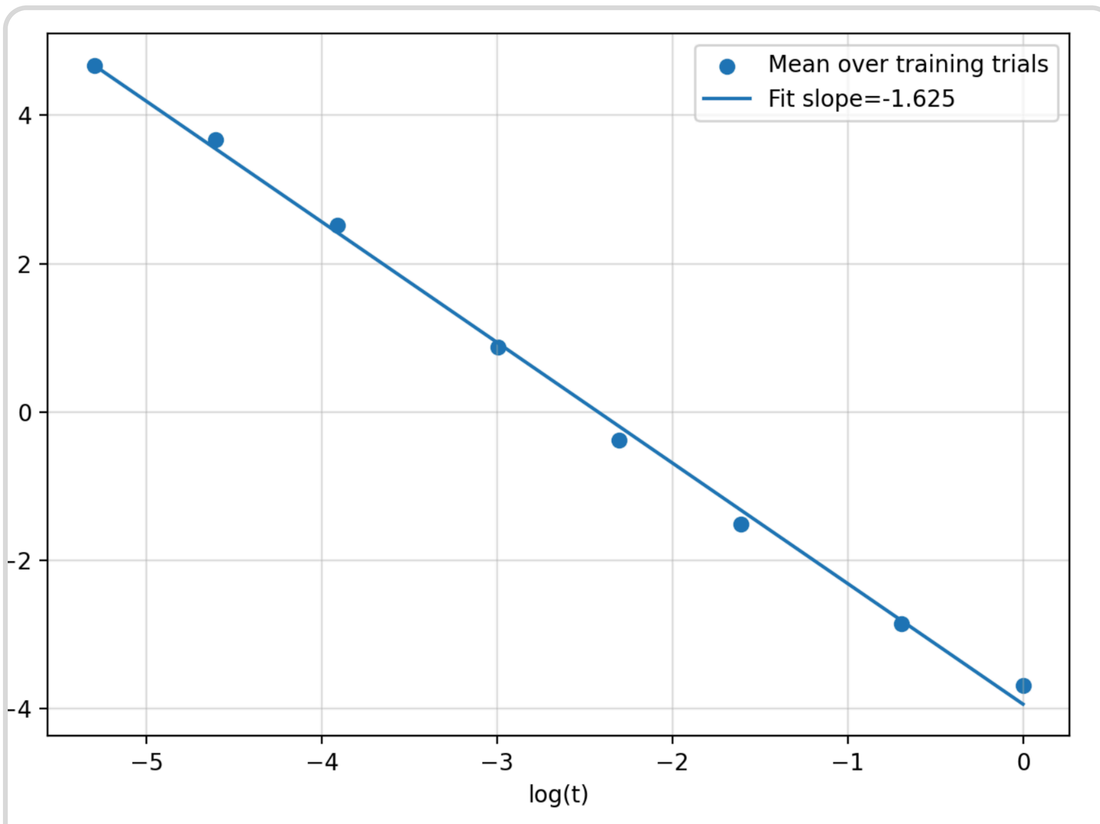
Independent ICML 2026 Reproduction Challenge audit

Public code & evidence

1 Question and scope

Can diffusion-model learning rates depend on intrinsic subspace dimension rather than ambient dimension? We report current local CPU experiments and explicit controls; no final claim verdict is asserted.

2 Source figures and rate



ure 1. Empirical L^2-score error versus diffusion time t.

Paper Figure 1: source score-error decay on the synthetic union-of-subspaces setting.

Finite controls currently do not establish an ambient-dimension-rate verdict.

4 Current outcomes

- **C1.** Inconclusive reverse-sampling test.
- **C2.** Inconclusive score-error controls.
- **C3.** Inconclusive assumption conditions.

C1–C4 are inconclusive. C5 is a reviewed 1-point d=1 Cai–Li toy, not a theorem verdict.

Evidence boundary. Source, controls, and SHA-256 manifests are retained; only C5 has a reviewed toy outcome.

Scope. The retained evidence includes finite local experiments but is insufficient for verified/falsified claims.

5 Assumption controls

C4 is inconclusive. C5 is a reviewed 1-point d=1 Cai–Li toy, not a theorem verdict.

into (26) yields

$$\sum_{j=0}^{L-1} \sqrt{(1-e^{-4T_j}) \int_{T_j}^{T_{j+1}} \mathbb{E}\big[\|\widehat{s}_{X_t}(X) - s^\star_{X_t}(X)\|_2^2\big]\, \mathrm{d}t}$$
$$\lesssim \sqrt{d}\, M^{3/2}\, \text{poly}\log n \begin{cases} \frac{1}{\sqrt{n}} \tau^{-\frac{k}{4}+\frac{1}{2}}, & k \geq 2 \\ \frac{1}{\sqrt{n}}, & k = 1 \end{cases}$$

Source-pinned page-9 rate excerpt; an equation display, not a second empirical result.

6 CPU reproduction route

1. Pin the paper PDF and arXiv source archive.
2. Implement local finite experiments and controls.
3. Record raw results, hashes, and limitations.
4. Do not self-award claim points.

7 Limitations

No author executable was available. Full neural diffusion training was not run; this logbook reports limited clean-room local evidence.

8 Takeaway

Claims 1–4 are inconclusive. C5 is a reviewed 1-point toy, not a rate/theorem conclusion.

- **C1–C2:** local sampling/score experiments remain inconclusive.
- **C3:** assumption diagnostics are not a full theorem derivation.

Reproducibility. Public code, source hashes, commands, and controls are retained; no official score is asserted.

Release contents. Contract text, source archive, claim artifacts, and deterministic tests are public and hash-addressed.

Interpretation. This is limited clean-room evidence, not a theorem verification or diffusion-training benchmark.